MIL-I-14811(MU)
21 October 1968

Superseding
FA(FC)PD-74
9 May 1961

MILITARY SPECIFICATION

INDICATOR, DIGITAL DISPLAY: 8215501

1. SCOPE

1.1 This specification covers one type of digital display indicator used in the Radar Chronograph M36. The assembly receives the detected doppler frequency signal, and converts it into projectile velocity which is indicated visually on five direct-reading neon display tubes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Processes and Inspection of
MIL-M-60673	Modules, Radar Chronograph: M36
MIL-M-60817	Modules, Radar Chronograph: M36 (Encapsulated)

FSC 1290

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

8215501	Indicator, Digital Display: 8215501

PACKAGING DATA SHEET

8215501	Packaging of Indicator, Digital Display

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The indicator shall be manufactured in accordance with Drawing F8215501 and drawings pertaining thereto.

3.2 General Specifications.- The indicator shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

3.2.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Etched circuit assemblies

3.3 Subassemblies.- Each indicator submitted in accordance with this specification shall incorporate subassemblies which have complied with all their respective requirements in accordance with the following specifications:

a. MIL-M-60673

Frequency Divider (Binary): 10525361
Frequency Divider (Decade): 10525362
Frequency Divider (Decade, Tens, and Thousands): 8215502
Indicator, Digital Display: 10558976

b. MIL-M-60817

Oscillator 75KC, Radio Frequency: 8215497
Oscillator 95.648KC, Radio Frequency: 8215500
Oscillator Doppler Gate: 10525492
Generator, Gate: 10525493
Pulse Generator: 10525494
Amplifier, Af/RF: 10525495
Indicator, Reset: 10525496
Gate Delay: 10525498

3.4 Environmental.

3.4.1 Shock.- The indicator shall be capable of continuous operation while being subjected to three (3) shocks of 50 gravity units (g's) of Eleven (11) millisecond duration in each of the following directions:

a. Bottom down
b. Front down
c. Right side down
d. Back down
e. Left side down
f. Top down

During the above shock, the indicator shall be reset (reset lamp lit) and display all zeroes continuously at all times. Subsequent to the above shock, the indicator shall not have been damaged or adversely affected, and shall meet the requirements of 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

3.4.2 Vibration.- The indicator shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from 10 to 55 hertz and return to 10 hertz, shall be uniformly traversed in one (1) minute ± 15 seconds. This motion shall be applied for a period of thirty (30) minutes ± one (1) minute in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the indicator shall meet the requirements of 3.6.4 once in each plane. At all other times, the indicator shall be in reset condition, displaying zeroes at all times. Subsequent to the above vibration, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.3 Humidity.- The indicator shall be capable of continuous operation during four (4) hours ± ten (10) minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) percent at a temperature of plus 100°F ± 5°F. During and subsequent to the above humidity, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.4 Sand and Dust.- The indicator shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm and blowing dust of particle size 0.0001 to 0.01mm with a density between 0.1 and 0.5 gram per cubic foot of dust at a temperature of plus 75°F ± 5°F, a relative humidity between fifteen (15) to thirty (30) percent, and a wind velocity of fifteen (15) knots. Subsequent to the above sand and dust exposure, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.5 Altitude

3.4.5.1 Operating.- The indicator shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F. During and subsequent to the above exposure, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.5.2 Non-Operating.- The indicator shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F. Subsequent to the above exposure, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.6 Fungus.- The indicator shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

a. Chaetomium globosum
b. Aspergillus flavus
c. Aspergillus versicalor
d. Penicillum luteum
e. Phizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 81°F ± 4°F with an RH of 87 ± 2%. Subsequent to the above exposure, the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6.

3.4.7 Storage temperatures.- The indicator shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight (8) hours at each temperature extreme. Subsequent to each of the above temperature extremes (+160°F and -80°F) after temperature stabilization at +75°F ± 15°, the indicator shall not have been damaged or adversely affected, and shall meet the requirements of 3.6.

3.4.8 Operating Temperatures.- The indicator shall be capable of continuous operation for eight (8) hours at +140°F. The indicator shall also be capable of continuous operation for eight (8) hours at minus 50°F. During and subsequent to the above temperature extremes (+140°F and -50°F), the indicator shall not have been damaged or adversely affected and shall meet the requirements of 3.6 at least once at the beginning, middle, and ending of each eight (8) hour temperature extreme. There shall be a minimum of five (5) to six (6) minutes warm-up time prior to performing the first test of 3.6 at -50°F.

3.5 Electrical

3.5.1 Continuity.- The indicator shall demonstrate continuity in accordance with the electrical schematic, Drawing F10525617.

3.6 Performance.- Unless otherwise specified, the indicator shall be capable of meeting the performance requirements of this specification at standard ambient temperatures between 60°F and 90°F and at a barometric pressure between twenty-eight (28) and thirty-two (32) inches of mercury, with an RH up to 90%. The performance requirements shall be capable of being met over a voltage range of +21.5 volts to +22.5 volts and +175 volts to +182 volts unless otherwise specified.

3.6.1 Reset.- When the reset switch is pressed, the indicator with no signal input, shall read 0000.0, the reset lamp shall light and remain lit until the indicator is triggered. When triggered, the reset lamp shall go out and the indicator shall not count and the nixie display shall read 0000.0.

3.6.2 Calibrate.- The internal calibration output, P1-8, shall be zero when the front panel OPER switch is in the center position. In CAL 2 and CAL 1 position, the output, P1-8, shall be a minimum of 30MV peak-to-peak sine wave at a frequency of 75KHz ± .012%.

3.6.3 Gate Delay

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

3.6.3.1 Room Temperature.- The gate delay before the start of the counting interval shall be 172 ± 3 msec with the delay switch in position 1, and shall increase by 42.82 ± 0.01 msec for each consecutive delay position when operating at a temperature of 75°F ± 15°F.

3.6.3.2 Extreme Temperatures.- The gate delay before the start of the counting interval shall be 172 ± 4 msec with the delay switch in position 1 and shall increase by 42.82 ± 0.01 msec for each consecutive delay position when operating over a temperature range of -50°F to +140°F.

3.6.4 Indicator Accuracy.- With the indicator reset and a 12KHz ± 0.01% 100MV peak-to-peak sine wave applied to the doppler input, P1-4, a nixie readout of 0171.3 shall be obtained when OPER switch is in CAL 2 position. Repeat above readout three times. The above shall be repeated using 110KHz ± 0.01% with a nixie readout of 1570.0, in place of 12KHz ± 0.01%. At -50°F, the inputs to P1-4 shall be increased to 150MV P-P to obtain the above readings.

3.6.5 Microphone Sensitivity.-

3.6.5.1 Microphone Sensitivity Switch At Zero Position.- With the microphone sensitivity switch in zero position and the indicator in reset condition (reset lamp lit), the reset lamp shall not go out when OPER switch is in CAL 2 position.

3.6.5.2 Microphone Sensitivity Switch in Position 5 through 10.- With the microphone sensitivity switch in positions 5 through 10 and the indicator in reset condition (reset lamp lit) the reset light shall go out when OPER switch is in CAL 2 position.

3.6.5.3 Microphone Input.- With the counter in reset condition (reset lamp lit), a square wave input applied to P1-1 with a maximum amplitude of 200mv at a frequency of 1KHz shall extinguish the reset light. The above performance shall be repeated using 10KHz in place of 1KHz.

3.6.6 Signal Meter Circuit.- With a 300mv peak-to-peak 75KHz sine wave applied to the doppler input a minimum output of 40ua shall be obtained at P1-10.

3.6.7 Test Points.- The following signals shall be obtained at the following test points:

a. J1.- With a 15KHz 500mv peak-to-peak sine wave applied to P1-1, the output shall be a square wave swinging between the levels of +2.3 ± 1.3 volts to +19.5 ± 1.5 volt.

b. J2.- With the indicator reset, the output at J2 shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

change levels from +0.5 ± 0.5 volts to +15.0 ± 1.8 volts and remain there for 86 ± 3 milliseconds and then return to +0.5 ± 0.5 volts, when OPER switch is in CAL 2 position.

c. J3.- The output at J3 shall be a 95.648KHz ± 0.012% square wave swinging between the levels of +11.5 ± 0.7 volts to a minimum negative change of 5 volts. At -50°F, the output shall swing between the levels of +11.5 ± 0.7 volts to a minimum negative change of 3.6 volts.

d. J4.- With the indicator reset, the output waveform at J4 shall be a 5.978 ± 0.12%KHz square wave swinging between the levels of +6.0 ± 0.8 volts to +11.5 ± 0.7 volts when OPER switch is in CAL 2 position. The duration of 5.978KHz output in delay 1 shall be 128.3 ± 0.1 msec and increase by 42.82 ± 0.01 msec for consecutive delay.

e. J5.- With the indicator reset, the output at J5 shall be a 373.6Hz ± 0.012% square wave swinging between the levels of +6.0 ± 0.8 to +11.5 ± 0.7 volts, when OPER switch is in CAL 2 position. The duration of the 373.6Hz output shall be 125.8 ± 0.1 msec in delay 1 and increase by 42.82 ± 0.01 msec, for each consecutive delay.

f. J6.- With the indicator reset, the output level at J6 shall change from +11.5 ± 0.7 volts to +6.0 ± 0.8 volts when OPER switch is in CAL 2 position, and remain there for 128.5 ± 0.1 msec in delay 1 and increase by 42.82 ± 0.01msec for each consecutive delay.

g. J7.- With the indicator reset, the output waveform at J7 shall be a 95.64KHz ± 0.012% square wave swinging between the levels of 6.0 ± 0.8 to +11.5 ± 0.7 volts when OPER switch is in CAL 2 position. The duration of the 95.648KHz output shall be 128.5 ± 0.1 msec in delay 1 and increase by 42.82 ± 0.01 msec for each consecutive delay.

h. J8.- With the indicator reset, the output waveform at J8 shall be a 23.35Hz ± 0.012% square wave swinging between the levels of +6.0 ± 0.8 to +11.5 ± 0.7 volts when OPER switch is in CAL 2 position. The duration of the 23.35Hz shall be 85.64 ± 0.1 msec in delay 1 and shall increase by 42.82 ± 0.01 msec for each consecutive delay.

i. J9.- With a 16KHz 100mv peak-to-peak sine wave applied to P1-4, the output at J9 shall be a 16KHz sine wave swing between the levels of +11.5 ± 0.7 volts to a minimum negative change of 3.6 volts. At -50°F, the input shall be increased to 150MV peak-to-peak to obtain the above results.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

j. J10.- With the indicator reset and a 16KHz 150mv peak-to-peak sine wave applied to P1-4, the output at J10 shall be a 16KHz square wave swinging between the levels of +5.6 ± 1.3 volts to +11.5 ± 0.7 volts when OPER switch is in CAL 2 position. The duration of the 16KHz output at J10 shall be 42.82 ± 0.01 msec.

k. J11.- With the indicator reset, the level at J11 shall change from +11.5 ± 0.7 volts to +5.6 ± 1.3 volts and back to 11.5 ± 0.7 volts when OPER switch is in CAL 2 position. The duration of the +5.6 volt level shall be 42.82 ± 0.01 msec.

l. J12.- Ground

m. J14.- With the indicator reset, the output at J14 shall be +6.0 ± 0.8 volts. When OPER switch is in CAL 2 position the output level at J14 shall switch to 11.5 ± 0.7 volts in delays 6 thru 10.

3.6.8 Nixie Display Indicator.- Each nixie indicator tube shall be visually monitored by injecting an audio square wave with an amplitude of 6.0 ± 0.1 volts into J10. The nixie indicator tubes and associated input frequencies are listed in Table I. Each number of the nixie indicator tubes shall light one at a time in a numerically progressive order. Each character (number) shall be 100% lit.

TABLE I

Input Frequency (J10)	Nixie Tube
1 Hz	Tenths
4 Hz	Units
40 Hz	Tens
400 Hz	Hundreds
4KHz	Thousands

NOTE: All numbers in units, tens and hundreds shall light. In tenths, only 0, 3, and 7 and in thousands, only 0 and 1 shall light.

3.7 Reliability Run-In.- In order to maximize component reliability, prior to acceptance by the Government, each indicator shall be subject to the following run-in conditions:

a. Temperature Run-In.- The indicator without power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, power shall be applied and each unit shall be tested to all the performance requirements of 3.6 for a period of three (3) hours with a minimum of one (1) complete test

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

to the requirements of 3.6 occurring at the beginning, middle and end of the three (3) hour period. Subsequent to the above tests, the indicator shall not have been damaged or adversely affected, and shall meet the requirements of 3.6. There shall be a minimum of five (5) to six(6) minutes warm-up time prior to the first performance test at -50°F.

b. Vibration Run-In.- The indicator shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz and return to ten (10) hertz, and shall be uniformly traversed in one (1) minute. This motion shall be applied for a period of ten (10) minutes in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 0.5 hours. During the above vibration, the indicator shall meet the requirements of 3.6.4 once in each plane. At all other times, the indicator shall be in rest condition and read zeroes at all times. During and subsequent to the above vibration, the indicator shall not be damaged or adversely affected, and shall meet the requirements of 3.6.

c. Shock Run-In.- The indicator shall be capable of continuous operation while being subjected to three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions: a. Bottom down; b. Front down; c. Right side down; d. Back down; e. Left side down; f. Top down.

During the above shock, the indicator shall be reset (reset lamp lit) and display all zeroes continuously at all times. Subsequent to the above shock, the indicator shall not have been damaged or adversely affected, and shall meet the requirements of 3.6.

3.7.1 Failures During Reliability Run-In.- Replacement of defective components and correction of workmanship defects may be made during run-in under the following conditions:

a. Any part, module, component or subassembly that is used as a replacement shall have previously passed environmental tests consistent with the reliability run-in environmental requirements of the indicator.

b. Any replacement part shall have been exposed for at least one (1) hour at the extreme temperature being run immediately previous to being used as a replacement.

c. After a correction has been accomplished, at least one (1) hour of operation with a minimum of one (1) test to the requirements of 3.6 shall be successfully accomplished at the temperature at which the failure occurred.

d. A failure during 3.7b or 3.7c shall necessitate a complete re-run of the conditions of 3.7b and 3.7c.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (Initial Production) Approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) indicators manufactured in the same manner using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions

4.3.1. Examination and Tests

4.3.1.1 Components and Subassemblies.- All components and subassemblies not covered by 3.3 shall be inspected in accordance with the inspection provisions contained in the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2 Acceptance and Rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

4.4 Special Sampling.- At the beginning of production and subsequent to First Article Approval (See 4.2), three units shall be selected by a Government representative and two from each fifty (50) produced. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401.	Fabrication	3.1	Applic Dwgs & Specs - Visual
402.	General Specifications	3.2	MIL-F-13926 & MIL-C-46357
403.	Shock	3.4.1	4.6.1.1
404.	Vibration	3.4.2	4.6.1.2
405.	Humidity	3.4.3	4.6.1.3
406.	Sand and Dust	3.4.4	4.6.1.4
407.	Altitude-Operating	3.4.5.1	4.6.1.5
408.	Altitude-Non-Operating	3.4.5.2	4.6.1.5.1
409.	Storage Temperatures	3.4.7	4.6.1.7.1
410.	Operating Temperatures	3.4.8	4.6.1.7.2

4.4.1 Failure of Sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by theGovernment until necessary corrections have been made by the contractor, and resubmitted samples have been approved (See 4.3.2).

4.5 Inspection Equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government Furnished Inspection Equipment.- Where the contract provides for Government-Furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

4.5.2.1 Contractor Design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, aterials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods and Procedures

4.6.1 Environmental.- Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities" requirements of MIL-F-13926 and conditions specified in 3.4.

4.6.1.1 Shock.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.4.1. Subsequent to the eighteen (18) shocks, the indicator shall be examined for physical damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.4.2. Subsequent to the 1.5 hour vibration, the indicator shall be examined for physical damage, then tested as specified in 4.6.2. During the above vibration, the indicator shall meet the performance tests as specified in 4.6.2.2.

4.6.1.3 Humidity.- The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions specified in 3.4.3. Subsequent to this test, the indicator shall be examined for damage then tested as specified in 4.6.2.

4.6.1.4 Sand and Dust.- The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.4.4. Subsequent to this test, the indicator shall be examined for damage then tested as specified in 4.6.2.

4.6.1.5 Altitude-Operating.- The altitude test shall be performed as specified in MIL-F-13926, Procedure I at an altitude of 10,000 feet

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

and as specified in 3.4.5.1. The indicator shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mercury) altitude condition. While the indicator is maintained at this pressure for four (4) hours, the indicator shall be subjected to the tests specified in 4.6.2.

4.6.1.5.1 Altitude Non-Operating.- The indicator shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.4.5.2. Subsequent to this test, the indicator shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.6 Fungus.- One (1) indicator shall be subjected to the fungus test specified in MIL-F-13926. At the end of the 28 day incubation period, the indicator shall be tested as specified in 4.6.2. for compliance with 3.4.6. Subsequent acceptance of indicators shall be by certification.

4.6.1.7 Temperature Cycling

4.6.1.7.1 Storage Temperatures.- This test shall be performed as specified in MIL-F-13926, Procedure II, and thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The indicator shall be returned to room ambient temperature+75° ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.1.7.2 Operating Temperatures.- The indicator shall be placed in a test chamber and the temperature gradually raised until thermally stabilized at +140°F for eight (8) hours. The indicator shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period, and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the indicator shall be tested as specified in 4.6.2 at the three (3) time periods specified above. A five (5) to six (6) minute warm-up period shall be allowed prior to performance of the first test. The indicator shall be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be performed on a 100% basis. All performance tests shall be met with the following inputs unless otherwise specified:

INPUTS:
a. +175VDC to pin 6 of P1 and +21.5VDC to pin 5 of P1
b. +182VDC to Pin 6 of P1 and +22.5 VDC to Pin 5 of P1
c. Ground to pin 7 of P1
d. Microphone sensitivity switch in position 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

4.6.2.1 Gate Delay

4.6.2.1.1 Room Temperature.- Connect the time interval counter as shown in Figure 1. Reset the indicator and depress CAL 2. The gate delay before the start of the counting interval shall be 172 ± 3 msec with the delay switch in position I, and shall increase by 42.82 ± 0.1 msec for each consecutive delay position when operating at a temperature of+75 ± 15°F.

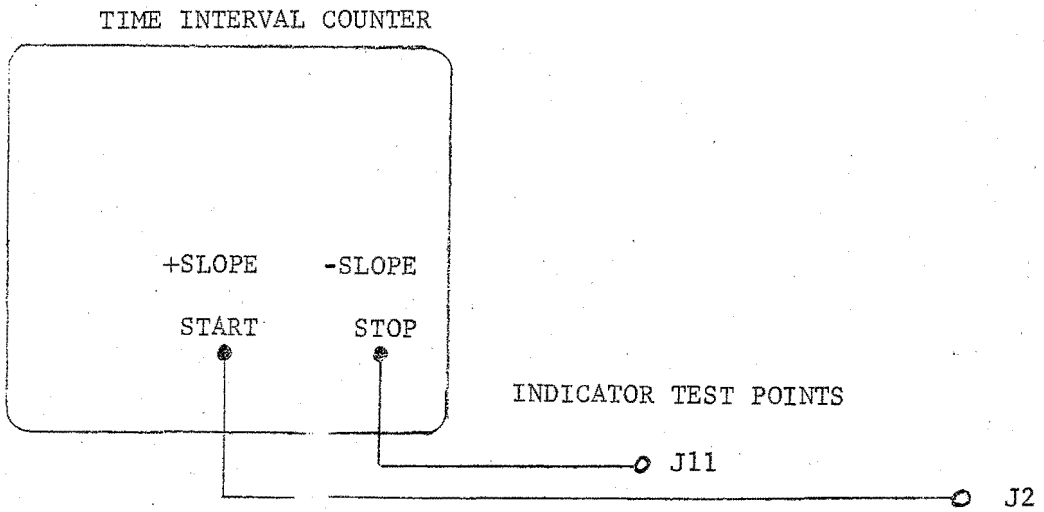


FIGURE I

4.6.2.1.2 Extreme Temperatures.- At -50 ± 3°F and +140° ± 3°F, the delay time shall be 172 ± 4 msec with the delay switch in position I, and shall increase by 42.82 ± 0.1 msec for each consecutive delay position.

4.6.2.2 Indicator Accuracy.- Press reset switch, reset lamp shall light. With a signal generator apply 100mv peak-to-peak sine wave at a frequency of 12KHz ± 0.01% to the doppler input P1-4. A nixie readout of 0171.3 $^{+0.4}_{-0.3}$ shall be obtained when OPER switch is placed to CAL 2 position. The above test shall be repeated at a frequency of 110KHz, with a nixie readout of 1570.0 ± 0.3. These readings shall be repeated in all delays 1 through 10. At -50 ± 3°F, the inputs to P1-4 shall be increased to 150MV peak-to-peak sine to obtain the readings in the above test.

4.6.2.3 Microphone Sensitivity

4.6.2.3.(a) Microphone Sensitivity Switch at Zero Position.- Place the microphone sensitivity switch in position zero. Press reset switch lighting reset lamp. Place OPER switch to CAL 2 position, the reset lamp shall not go out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

4.6.2.3.(b) Microphone Sensitivity Switch in Positions 5 through 10.- Place the microphone sensitivity switch in position 5. Press reset switch lighting reset lamp. Place OPER switch to CAL 2 position, the reset lamp shall go out. The above test shall be repeated with the microphone sensitivity switch in positions 6 through 10.

4.6.2.4 Microphone Input.- Place the microphone sensitivity switch in position 10 and the reset lamp lit (press reset switch). With a signal generator, apply a 50mv peak-to-peak square wave at a frequency of 1KHz to P1-1. By increasing the input to P1-1, the reset lamp shall be extinguished at a value less than 200mv peak-to-peak. The above test shall be repeated using 10KHz in place of 1KHz.

4.6.2.5 Signal Meter Circuit.- Apply 300mv peak-to-peak 75KHz sine wave to P1-4. A minimum output of 40ua shall be obtained at P1-10 as indicated by an ammeter. (Ammeter internal impedance shall be 1.1K Ohms ± 5%.)

4.6.2.6 Test Points.- The output signals shall be obtained at the following test points and ground (j-12).

a. J1.- With a signal generator, apply a 15KHz 500mv peak-to-peak sine wave to P1-1. The output at J1 shall be a square wave swinging between the levels of +2.3 ± 1.3 volts to +19.5 ± 1.5 volts, as indicated on the oscilloscope.

b. J2.- Connect an oscilloscope and time interval counter to J2. Press reset switch, reset lamp shall light. The output at J2 shall change levels from +0.5 ± 0.5 to +15.0 ± 1.8 volts and remain there for 86 ± 3 milliseconds and return to +0.5 ± 0.5 volts, when OPER switch is placed to CAL 2 position.

c. J3.- Connect the oscilloscope to J3. The output at J3 shall be a 95.648KHz ± 0.012% square wave swinging between the levels of +11.5 ± 0.7 volts to a minimum negative change of 5 volts. At -50°F, the output shall swing between the levels of +11.5 ± 0.7 volts to a minimum negative change of 3.6 volts.

d. J4.- Connect the oscilloscope to J4. Press reset switch, reset lamp shall light. The output waveform at J4 shall be a 5.978 ± 0.12%KHz square wave swinging between the levels of +6.0 ± 0.8 volts to +11.5 ± 0.7 volts when OPER switch is placed to CAL 2 position. The duration of 5.978KHz output in delay 1 shall be 128.3 ± 0.1 msec and shall progressively increase by 42.82 msec for each consecutive delay (Duration of 5.978KHz is equal to 85.5 +n 42.82 ± 0.1 msec where n = delay 1, delay 2, through delay 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

e. J5.- Connect the oscilloscope to J5. Press reset switch, reset lamp shall light. The output at J5 shall be a 373.6Hz ± 0.012% square wave swinging between the levels of +6.0 ± 0.8 volts to +11.5 ± 0.7 volts, when OPER switch is placed to CAL 2 position. The duration of the 373.6Hz square wave output in delay 1 shall be 125.8 ± 0.1 msec and shall progressively increase by 42.82 msec for each consecutive delay (Duration of 373.6Hz square wave is equal to 82.98 +n 42.82 ± 0.1 msec where n = delay 1, delay 2, through delay 10.

f. J6.- Connect the oscilloscope and time interval counter to J6. Press reset switch, reset lamp shall light. The output level at J6 shall change from 11.5 ± 0.7 volts to +6.0 ± 0.8 volts when OPER switch is in CAL 2 position and remain there for 128.5 ± 0.1 msec and shall progressively increase by 42.82 for each consecutive delay (Duration of +6.0 ± 0.8 level is equal to 85.68 +n 42.82 ± 0.1 msec where n = delay 1, delay 2, through delay 10.)

g. J7.- Connect the oscilloscope to J7. Press reset switch, reset lamp shall light. The output level at J7 shall be a 95.648KHz ± 0.012% square wave swinging between the levels of +11.5 ± 0.7 to +6.0 ± 0.8 volts when OPER switch is placed to CAL 2 position. The duration of the 95.648KHz output shall be 128.5 ± 0.1 and shall progressively increase by 42.82 msec for each consecutive delay (Duration of 95.648KHz square wave shall be equal to 85.6 +n 42.82 ± 0.1 msec where n = delay 1, delay 2, through delay 10.)

h. J8.- Connect the oscilloscope to J8. Press reset switch, reset lamp shall light. The output wave form at J8 shall be a 23.35Hz ± 0.012% square wave swinging between the levels of +6.0 ± 0.8 volts to +11.5 ± 0.7 volts when OPER switch is placed to CAL 2 position. The duration of the 23.35Hz square wave shall be 85.64 ± 0.1 msec and shall progressively increase by 42.82 msec for each consecutive delay (Duration of 23.35Hz square wave is equal to 42.82 +n 42.82 ± 0.1 msec where n = delay 1, delay 2, through delay 10).

i. J9.- With the signal generator, apply a 16KHz 100mv peak-to-peak sine wave to PI-4. The oscilloscope at J9 shall display a 16KHz sine wave as shown in Figure 2. At -50°F, the above input shall be increased to 150MV peak-to-peak to obtain the above results.

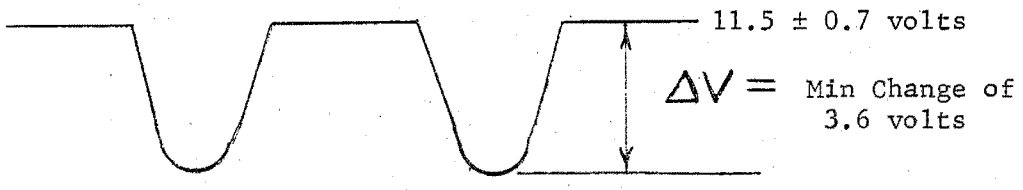


FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

j. J10.- Connect the oscilloscope to J10. Press reset switch, reset lamp shall light. With the signal generator, apply a 16KHz 150mv peak-to-peak sine wave to PI-4. The output at J10 shall be a 16KHz square wave swinging between the levels of 5.6 ± 1.3 volts to 11.5 ± 0.7 volts when OPER switch is placed to CAL 2 position. The duration of the 16KHz square wave is 42.82 ± 0.01 msec.

k. J11.- Connect the oscilloscope and time interval counter to J11. Press reset switch, reset lamp shall light. The output at J11 shall change from +11.5 ± 0.7 volts to +5.6 ± 1.3 volts and back to +11.5 ± 0.7 volts when OPER switch is placed to CAL 2 position. The duration of the +5.6 volt level shall be 42.82 ± 0.01 msec.

1. J14.- Connect the oscilloscope to J14. Press reset switch, reset lamp shall light. The output level at J14 shall be +6.0 ± 0.8 volts. When OPER switch is placed to CAL 2 position, the above output level shall switch to +11.5 ± 0.7 volts in delays 6 through 10.

4.6.2.7 Nixie Display Indicator.- Each nixie indicator tube shall be visually monitored by injecting an audio square wave with an amplitude of 6.0 ± 0.1 volts into J10. Insert a .047 uf capacitor between J10 and audio generator. The nixie indicator tubes and associated monitoring frequencies are listed in Table I. Each number of the nixie indicator tubes shall light (100%) one at a time in a numerically progressive order.

4.6.3 Electrical

4.6.3.1 Continuity.- Continuity shall be tested by standard measuring equipment for compliance with 3.4.1.

4.6.4 Reliability Run-In.- The following run-in tests shall be performed on a 100% basis.

4.6.4.1 Temperature Run-In.- The indicator shall be placed in a chamber and the temperature gradually raised until thermally stabilized at plus 125°F. After temperature stabilization, power shall be applied and maintained for three (3) hours. Five (5) minutes after power is applied, a complete test to the requirements of paragraph 4.6.2 shall commence. A second test to the requirements of paragraph 4.6.2 shall be accomplished during the second hour and a third test to the requirements of paragraph 4.6.2 shall be accomplished during the third hour of the three (3) hour test. Repeat the above tests at a temperature of minus 25°F. Subsequent to the above temperature extremes (plus 125°F and minus 25°F) the indicator shall not have been damaged or adversely affected and shall meet the requirements of paragraph 4.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

4.6.4.2 Vibration Run-In.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.7b. During the above vibration test, the indicator shall meet the performance tests as specified in paragraph 4.6.2.2 and 4.6.2.1.1. Subsequent to the vibration test, the indicator shall be visually examined for damage and shall meet the performance requirements of paragraph 4.6.2..

4.6.4.3 Shock Run-In.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.4.1. Subsequent to the eighteen (18) shocks, the indicator shall be visually examined for physical damage, then tested as specified in 4.6.2.

4.6.4.4 Failures.- Failures occurring during Reliability Run-In shall not be cause for rejection, but shall be corrected as outlined in paragraph 3.7.1.

4.6.5 Packaging, Packing, and Marking.- Quality Assurance for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

5. PREPARATION AND DELIVERY

5.1 Packaging, Packing, and Marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 8215501. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use.- The indicator is used as a subassembly of the Radar Chronograph M36, which gives a nixie display in meter/sec of the doppler return.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable stock number
d. Applicable packaging data sheet (See 5.1)

Custodian:

Army: MU

Preparing Activity:
Army: MU

Project: No. 1290-A199

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

343-221 (A-519)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:48Z
Check the source to verify that this is the current version before use.